



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-4	Team No-2	Project Title- Aircraft Maintenance Tracker
Team Lead Roll Number	2520090115	
Team Lead Name	SAI NATH REDDY	
Team Member Roll Number	-	
Team Member Name	-	

1. Abstract

The Aircraft Maintenance Tracker is an intelligent web-based system designed to manage, monitor, and streamline aircraft maintenance activities efficiently. The main objective of this system is to improve aviation safety, reduce maintenance delays, and maintain accurate records of aircraft inspections, repairs, spare parts, and technician assignments.

The system maintains aircraft details such as aircraft ID, model, last service date, flight hours, and maintenance status. It supports scheduled maintenance planning based on usage hours, calendar dates, and safety requirements. The system can generate alerts for upcoming inspections, overdue maintenance tasks, critical defects, and spare part shortages.

It also provides modules for defect reporting, technician allocation, spare parts inventory, maintenance history, compliance reports, and an admin dashboard. By automating maintenance tracking and reporting, the Aircraft Maintenance Tracker improves operational reliability, ensures timely servicing, and demonstrates the practical use of front-end frameworks, UI engineering, data management, and algorithmic thinking in the aviation domain.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Aircraft profile	2520090115	SAI NATH REDDY
Maintenance schedule	2520090115	SAI NATH REDDY
Inspection checklist	2520090115	SAI NATH REDDY
Defect report	2520090115	SAI NATH REDDY
Technician assignment	2520090115	SAI NATH REDDY
Spare parts inventory	2520090115	SAI NATH REDDY
Maintenance history	2520090115	SAI NATH REDDY
Alert notifications	2520090115	SAI NATH REDDY
Compliance reports	2520090115	SAI NATH REDDY
Admin panel	2520090115	SAI NATH REDDY



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3. Module Details

The Aircraft Maintenance Tracker system is divided into multiple integrated modules that collectively handle aircraft information, scheduled service planning, inspection checklists, defect reporting, and technician assignment. The aircraft profile module stores essential aircraft data, while the maintenance schedule module tracks upcoming and overdue services. The inspection checklist and defect report modules help record safety checks, faults, and repair status. The technician assignment module allocates maintenance tasks to available staff, and the spare parts inventory module monitors part availability and stock requirements. The system also maintains maintenance history for future reference, sends alert notifications for urgent activities, generates compliance reports, and provides an admin panel for centralized control of the entire aircraft maintenance process.

4. Frameworks / Platforms / Software Used

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Guide Name and Signature (remarks if any) : Dr. NEELIMA GOGINENI